

# ELEC 2FH4

---

## Tutorials

---

---

---

---



# Tutorial #1

1) unit vector from  $\vec{O}$  to  $\vec{G}(-2, -2, -1)$

$$\begin{aligned} |\vec{G}| &= \sqrt{2^2 + 2^2 + 1} \\ &= \sqrt{4 + 4 + 1} \\ &= \sqrt{9} \\ &= 3 \end{aligned}$$

$$\begin{aligned} \vec{a}_g &= \frac{1}{|\vec{G}|} \vec{G} \\ &= \left( \frac{-2}{3}, \frac{-2}{3}, \frac{-1}{3} \right) \\ &= \left( -\frac{2}{3} a_x, -\frac{2}{3} a_y, -\frac{1}{3} a_z \right) \end{aligned}$$

2) transform  $\vec{B}$  into cylindrical coords.  $\vec{B} = y a_x - x a_y + a_z$

rectangular  $\rightarrow$  cylindrical  
 $(x, y, z) \rightarrow (\rho, \theta, z)$

$$\begin{aligned} \rho &= \sqrt{x^2 + y^2} & \theta &= \tan^{-1} \left( \frac{y}{x} \right) & z &= z \\ &= \sqrt{y^2 + (-x)^2} & &= \tan \left( \frac{-x}{y} \right) & \end{aligned}$$

3)  $G = \left( \frac{xz}{y} \right) a_x \rightarrow$  spherical components

$$\begin{aligned} x &= r \sin \theta \cos \phi \\ y &= r \sin \theta \sin \phi \\ z &= r \cos \theta \end{aligned}$$

$$\begin{aligned} a_x &= \sin \theta \cos \phi a_r + \cos \theta \cos \phi a_\theta - \sin \phi a_\phi \\ a_y &= \sin \theta \sin \phi a_r + \cos \theta \sin \phi a_\theta + \cos \phi a_\phi \\ a_z &= \cos \theta a_r - \sin \theta a_\theta \end{aligned}$$

$$G = \left( \frac{\cancel{r \sin \theta} \cos \theta \cdot r \cos \theta}{\cancel{r \sin \theta} \sin \phi} \right) a_x$$

$$= (r \cot \phi \cos \theta) a_x$$

$$= (r \sin \theta \cos \theta \cot \phi) \vec{a}_r + (r \cos^2 \theta \cos \phi \cot \phi) a_\theta + (-r \sin \phi \cos \theta \cot \phi) a_\phi$$

4)  $\vec{D} = ?$  given  $P(10, 150^\circ, 330^\circ)$   
 $(r, \theta, \phi)$

$\vec{D}$  spherical coords?

5)  $\vec{D} = \underset{\substack{\uparrow \\ \text{normal}}}{D_n} + \underset{\substack{\uparrow \\ \text{tangential}}}{D_t}$

$$D_t = D - D_n$$

$D(p)$  - normal part of  $\vec{D}(p)$   
 $= -5a_r$

$$\vec{D}(p) - D_r \sin \theta = -5a_r$$

$$D_t = -0.043 a_\theta + 100 a_\phi$$

6)  $\vec{D} \times a_\theta = \begin{vmatrix} a_r & a_\theta & a_\phi \\ -5 & 0.043 & 100 \\ 0 & 1 & 0 \end{vmatrix}$

$$-100 a_r - 5 a_\phi$$

$$|\vec{a}| = \sqrt{100^2 + 5^2}$$

$$\hat{a} = \frac{\vec{a}}{|\vec{a}|} = -0.998 a_r - 0.0499 a_\phi$$

7)  $\vec{E} = -5a_r + 10a_\theta + 3a_z$

$$\vec{F} = a_r + 2a_\theta - 6a_z$$

$$\begin{aligned} \vec{E} \times \vec{F} &= \begin{vmatrix} 5 & 10 & 3 \\ 1 & 2 & -6 \end{vmatrix} = \begin{matrix} -66 a_\theta - 20 a_z - 33 a_r \\ \sqrt{(-66)^2 + (-20)^2 + (-33)^2} \end{matrix} = ? \\ &= 74.06 \end{aligned}$$